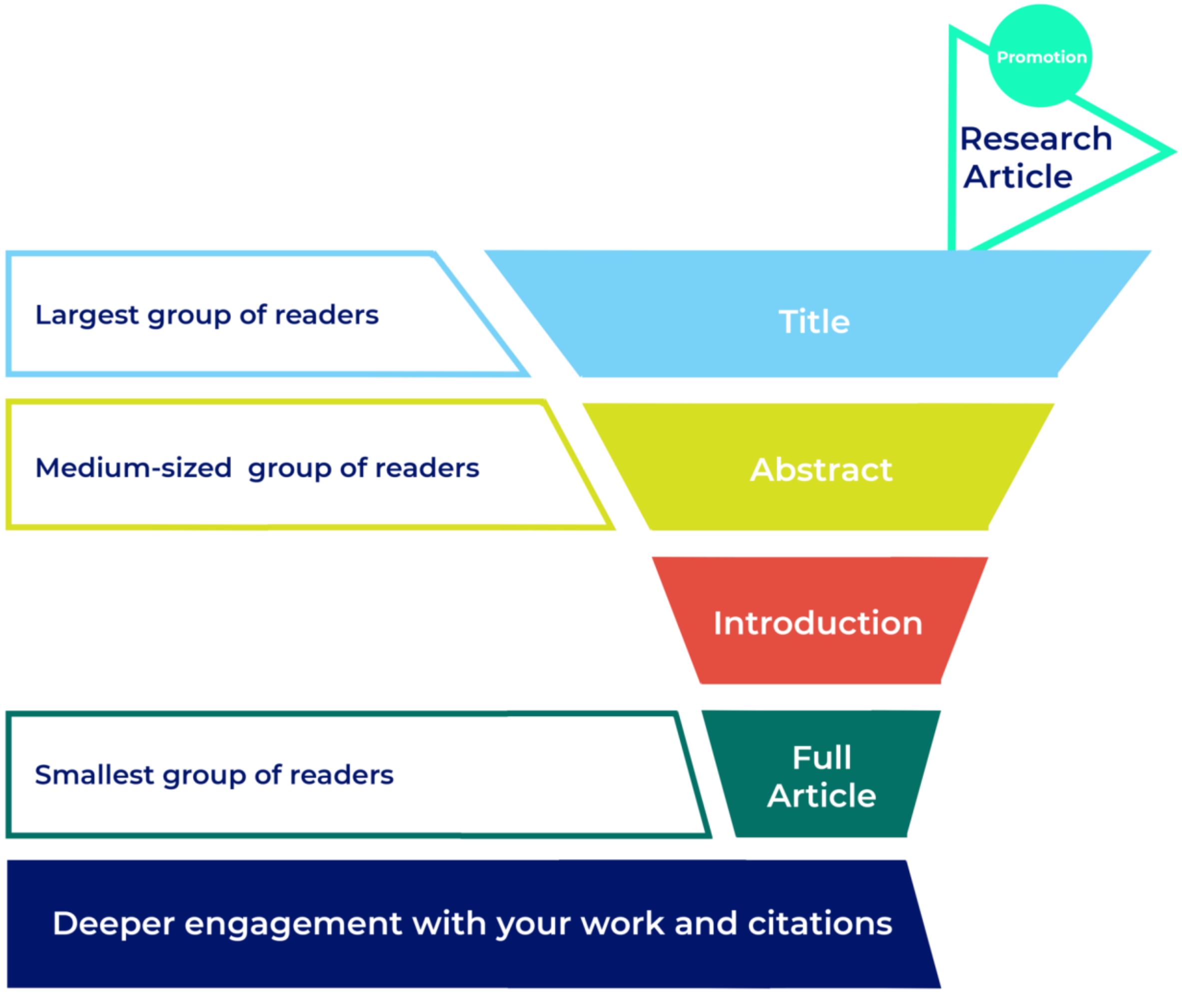
How to write an excellent title and abstract

How to write a great title

- The title is the first thing people read
- Help people find your research -> keywords
- Include:
  - Key information about the study
  - Important keywords
  - What you discovered



How to write a great title

Do:

- Keep it concise and state your main finding: Aim for fewer than 12 words
- Write for your audience: specialists, cross-disciplinary, or non-specialists?
- Entice the reader: Catch readers' interest, give information to keep them reading
- Include important keywords: Important for searching for your work
- Write in sentence case: Capitalize only the first word and proper nouns

Do not:

- Do not use waste words like "Studies on" or "Investigations on"
- Do not end your title with a question mark?
- Do not use abbreviations in a title

Examples

- Supply chain diversity protects cities against food shocks
- Effect of supply chain diversity on food shocks

- A highly magnetized and rapidly rotating white dwarf as small as the Moon
- Use of the Zwicky Transient Facility to search for short period objects below the main white dwarf cooling sequence

"**How to tell a compelling story in scientific presentations" – Point#1: State your main finding in your title**
https://www.nature.com/articles/d41586-021-03603-2

How to write an excellent abstract

Do:

- Keep it concise and to the point. Usually 150-300 words
- Write for your audience: specialists, cross-disciplinary, or non-specialists?
- Follow the standard format
- Include keywords or phrases to help readers find your work in PubMed / Google Scholar
- Double check for spelling and grammar errors

Do not:

- Do not speculate
- Do not use abbreviations or acronyms
- Do not repeat yourself unnecessarily, eg. “We used X technique. Using X technique, we found…”
- Do not include content that isn’t in the main manuscript
- Do not include citations

How to write an excellent abstract

Annotated example taken from *Nature* **435,** 114–118 (5 May 2005).

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

Two to three sentences of **more detailed background**, comprehensible to scientists in related disciplines.

One sentence clearly stating the **general problem** being addressed by this particular study.

One sentence summarizing the main result (with the words “**here we show**” or their equivalent).

Two or three sentences explaining what the **main result** reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge.

One or two sentences to put the results into a more **general context**.

Two or three sentences to provide a **broader perspective**, readily comprehensible to a scientist in any discipline, may be included in the first paragraph if the editor considers that the accessibility of the paper is significantly enhanced by their inclusion. Under these circumstances, the length of the paragraph can be up to 300 words. (This example is 190 words without the final section, and 250 words with it).

During cell division, mitotic spindles are assembled by microtubule-based motor proteins[1,2]. The bipolar organization of spindles is essential for proper segregation of chromosomes, and requires plus-end-directed homotetrameric motor proteins of the widely conserved kinesin-5 (BimC) family[3]. Hypotheses for bipolar spindle formation include the ‘push–pull mitotic muscle’ model, in which kinesin-5 and opposing motor proteins act between overlapping microtubules[2,4,5]. However, the precise roles of kinesin-5 during this process are unknown. Here we show that the vertebrate kinesin-5 Eg5 drives the sliding of microtubules depending on their relative orientation. We found in controlled *in vitro* assays that Eg5 has the remarkable capability of simultaneously moving at ~20 nm s^{-1} towards the plus-ends of each of the two microtubules it crosslinks. For anti-parallel microtubules, this results in relative sliding at ~40 nm s^{-1}, comparable to spindle pole separation rates *in vivo*[6]. Furthermore, we found that Eg5 can tether microtubule plus-ends, suggesting an additional microtubule-binding mode for Eg5. Our results demonstrate how members of the kinesin-5 family are likely to function in mitosis, pushing apart interpolar microtubules as well as recruiting microtubules into bundles that are subsequently polarized by relative sliding. We anticipate our assay to be a starting point for more sophisticated *in vitro* models of mitotic spindles. For example, the individual and combined action of multiple mitotic motors could be tested, including minus-end-directed motors opposing Eg5 motility. Furthermore, Eg5 inhibition is a major target of anti-cancer drug development, and a well-defined and quantitative assay for motor function will be relevant for such developments.

https://www.nature.com/nature/for-authors/formatting-guide

Wednesday writing workshop:
Peer evaluations of title and abstract